



OFFICIAL PROJECT DOCUMENTATION

UPSC Mentor Technical & Functional Architecture Report

A comprehensive specification for an AI-powered Civil Services examination preparation platform

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Overview

UPSC Mentor is a full-stack, AI-powered examination preparation platform purpose-built for aspirants of the Union Public Service Commission (UPSC) Civil Services Examination (CSE) — India's most competitive national examination. The platform consolidates every critical dimension of CSE preparation into a single, coherent workspace: structured syllabus tracking, real-time study timing, AI-graded Mains answer evaluation, Prelims PYQ drilling, timed mock testing with automated diagnostic reports, and a persistent multi-thread AI mentor with long-term student memory.

The platform is deployed at upscbyitiians.in and is architected on a React/Vite single-page application frontend, a Node.js/Express backend, PostgreSQL for persistent data storage, Socket.io for real-time cross-device synchronisation, and a multi-provider AI fallback chain (Google Gemini → Groq → OpenRouter) ensuring continuous availability of all AI-driven features.

Target Market

The primary audience is the estimated **900,000+ active UPSC CSE aspirants** in India who compete annually for approximately 1,000 posts across the Indian Administrative Service (IAS), Indian Police Service (IPS), Indian Foreign Service (IFS), and allied services. The examination spans three stages — Prelims, Mains, and Interview — across a preparation cycle of one to three years, creating sustained demand for structured, intelligent preparation tools.

Problems Addressed

Problem	Impact on Aspirants	UPSC Mentor Solution
Fragmented preparation ecosystem	Aspirants typically manage 5–8 separate applications for timers, syllabus tracking, MCQ practice, notes, and answer writing	A single unified dashboard integrating all preparation pillars with a shared data model
High cost of quality mentorship	Personalised coaching costs ₹50,000–₹2,00,000 per year, placing it beyond reach for most aspirants	AI Mentor workspace with persistent student memory, available 24/7 at no marginal cost
Delayed Mains answer feedback	Aspirants write practice answers with no structured feedback for days or weeks	Instant AI evaluation with score, keyword analysis, structural feedback, and a topper-level model answer
Absence of performance diagnostics	MCQ test scores are viewed in isolation; weak topics are not systematically identified or addressed	AI-generated diagnostic reports after every mock attempt with topic-wise breakdown and a personalised 7-day study plan
Inconsistent cross-device tracking	Study hours logged on one device are not reliably reflected across others	Real-time cross-device study timer synchronisation via WebSocket infrastructure

Platform Traction

UPSC Mentor has achieved meaningful early traction among serious UPSC aspirants, with confirmed usage by students from premier institutions including IITs and other top-ranked universities. The platform has established a committed early-adopter base whose behavioral patterns demonstrate strong product-market fit.

~60%

Next-Day Retention

Approximately 60% of users returned to the platform the day after their first session, indicating strong initial value delivery and product stickiness.

~40%

Multi-Feature Engagement

Approximately 40% of users engaged with multiple core platform features within their first few sessions, demonstrating cross-functional platform utility.

5+ IITs

Institutional Reach

Trusted and actively used by students from premier institutions including IITs, reflecting credibility among high-achieving, discerning aspirants.

Unified

Integrated Workflow Adoption

Users regularly combine study tracking, syllabus management, AI mentoring, and answer evaluation within a single cohesive preparation workflow.

PRODUCT-MARKET FIT SIGNAL

Early usage patterns — particularly the cross-feature engagement and next-day retention rates — indicate strong product-market fit among serious UPSC aspirants committed to structured, data-driven preparation. Qualitative feedback consistently highlights improvements in study consistency, answer writing quality, and examination confidence.

User Outcomes

Early adopter feedback and observable usage patterns reflect tangible preparation improvements across the platform's core workflows:

Improved Study Consistency

Daily tracking systems, streak-based accountability, and exam countdown visualisation have measurably improved adherence to study schedules among regular users.

Faster Weak Topic Identification

AI-powered diagnostic reports following each mock test attempt surface specific weak topics and generate targeted study plans, enabling faster course correction than manual review.

Consolidated Preparation Workflow

Users report significantly reduced reliance on multiple disconnected tools, with the integrated platform replacing separate applications for timers, PYQ banks, answer writing, and mentoring.

Accelerated Answer-Writing Development

Instant AI evaluation with rubric-based scoring, keyword diagnostics, and structured rewrites has dramatically shortened the feedback loop for Mains answer writing practice.

SECTION 03

The Functional Ecosystem — Core Features

Dashboard Engine

Cross-Device Subject Study Timer

The study timer operates as a persistent, subject-aware module spanning all **12 UPSC timer subjects**. It maintains accurate elapsed study time per subject across page navigation, browser tab switches, and device changes. On session start, it synchronises with the server's stored daily log to prevent double-counting, then broadcasts live state updates to all open tabs and devices via Socket.io. Periodic server-side persistence ensures no study time is lost on unexpected session termination. An IST-aware midnight rollover mechanism correctly segments daily logs across calendar day boundaries without requiring a scheduled background job.

Syllabus Sync Modal

When a subject study session ends, aspirants are presented with a confirm-before-updating checklist of the real Prelims/Mains syllabus modules mapped to that subject, split into switchable **Prelims / Mains tabs** with a remaining-topic count badge on each tab. Only the topics the student actively checks off are marked complete — nothing is auto-marked from elapsed time alone — with a free-text fallback note available for study that falls outside the mapped topic list. This mapping is powered by an exhaustive subject-to-syllabus-module table maintained independently for Prelims and Mains; modules with no clean 1:1 subject fit (e.g. Indian Society & Diversity, Internal Security, International Relations) are deliberately left unmapped rather than forced into an inaccurate bucket.

Subject Analytics Dashboard

Each of the 12 subjects rolls up into daily, weekly, monthly, and lifetime accumulation views, bridged directly to syllabus completion — every hour of logged study time contributes 10% toward that subject's displayed progress, capped at 95% so that the final push to full completion still requires the aspirant to explicitly confirm topic coverage through the Syllabus Sync Modal rather than time alone.

Streak & Countdown System

The platform maintains both a current consecutive-day streak and an all-time maximum streak per user, updated server-side on each authenticated interaction. The exam countdown is derived from the user's configured target exam date and refreshed on each dashboard load, providing a persistent sense of urgency and progress that supports long-term discipline.

Study Chart & Data Visualisation

The Study Chart renders a bar chart comparing actual hours studied against the user's personal daily target across weekly, monthly, and yearly views. Data is drawn from server-side daily log records, enabling reliable cross-device reporting and long-term trend analysis of preparation intensity.

Today Planner & Current Affairs Log

The daily planner provides a lightweight task management interface for setting and tracking daily preparation goals. A dedicated Current Affairs Log allows aspirants to capture and organise daily headlines, which is a critical input for both Prelims and Mains. Both features are synchronised to the backend, ensuring continuity across sessions and devices.

Answer Writing Tracker, Paper Progress & ActionHub

An Answer Writing Tracker logs Mains answer-writing practice performed outside the formal AI evaluator flow, capturing raw writing volume as a discipline metric in its own right. Paper Progress surfaces an at-a-glance completion percentage per GS paper, pulled directly from the Syllabus Tracker so aspirants always know which paper needs attention next. The ActionHub presents four quick-access cards for the platform's core study workflows, reducing the dashboard's cognitive load to a single next-action decision.

DASHBOARD AS COMMAND CENTRE

Together, the timer, syllabus sync, streak system, study chart, daily planner, and ActionHub are designed to work as a single connected surface rather than separate widgets — every session logged here immediately informs syllabus coverage, streak continuity, and the aspirant's next recommended action, so the Dashboard functions as the operational command centre for daily preparation.

Syllabus Tracker

The full UPSC CSE syllabus — covering Prelims GS Paper I, CSAT, and Mains GS1–GS4 and Essay — is provisioned as 48 individual modules per user on account creation, each derived directly from the official UPSC CSE notification. Each module stores its exam stage, paper classification, sub-topic list, coverage percentage, and preparation state (Pending / In Progress / Revision / Done).

A batch update architecture allows aspirants to update an entire paper's module states in a single server interaction, eliminating the latency and consistency risks of sequential per-module update patterns. This design makes large-scale syllabus reorganisation sessions fast and reliable.

SYLLABUS COVERAGE

The platform seeds **48 syllabus modules** per user on registration — spanning 7 Prelims GS modules, 7 CSAT modules, 7 GS1 modules, 7 GS2 modules, 6 GS3 modules, 8 GS4 modules, and 6 Essay modules — each derived directly from the official UPSC CSE notification.

AI Mentor Workspace

Persistent Multi-Thread Architecture

The AI Mentor supports a multi-conversation thread model, mirroring the interaction paradigm of modern AI assistants. Aspirants can create, rename, switch between, and delete independent conversation threads, allowing them to maintain separate lines of inquiry — for example, one thread for GS4 ethical case studies and another for economic policy analysis. Full conversation history is passed with each request, ensuring coherent, context-aware responses across multi-turn interactions.

Durable Student Memory

A background memory extraction process runs silently after each conversation turn, distilling a curated set of student-specific facts — weak topics, study patterns, exam goals, and recurring conceptual gaps — from the conversation history. This memory is injected into future prompts without being narrated back to the student, giving the AI Mentor persistent awareness of the student's preparation context across all sessions. Memory extraction failures are handled gracefully with no impact on the primary chat experience.

Structured Response Standards

The AI Mentor's conversational behaviour is governed by a detailed system instruction that enforces structured, substantive responses using headings, tables, and explicit memory-recall blocks when drawing on stored student facts. The instruction is designed to produce the behaviour of a demanding but supportive human mentor rather than a generic question-answering assistant.

Quote-of-the-Day Deployment

A 365-quote rotation engine surfaces a fresh quote card each hour on the landing hero. Any card can be sent directly into the AI Mentor chat for instant explanation, historical context, and guidance on deploying it as an opening line in an Essay or GS answer — turning a passive motivational element into an active writing-practice input.

Mentor Notes — AI-Powered Notes Workspace

A dedicated two-pane notes workspace pairs a note-list sidebar with a full editor, organised across **8 UPSC subject tags (Polity, History, Economy, Geography, Sociology, Ethics, Environment, Science & Tech)** for filtering. Four AI actions run on any note through the same multi-provider chain used elsewhere on the platform: Improve Notes, Find Mistakes (returning Knowledge, Clarity, and Retention scores), Generate Revision Notes, and Convert to Mains Format.

A rich reading mode renders headings, bullet lists, memory-card callouts, and [!NOTE] / [!TIP] / [!WARN] callout boxes, inline code, and tables from a lightweight markdown-lite syntax. Every note keeps an undo history of up to 40 content snapshots (2-second debounce, triggered by Ctrl/Cmd+Z or the History button), and a snapshot is pushed automatically before every AI rewrite so any AI edit is fully reversible. Content autosaves with a 1.4-second debounce and flushes on tab switch or unmount, with a relative-time save indicator. Notes are currently stored client-side; the AI actions themselves require sign-in, surfacing a sign-in gate drawer otherwise.

Mains Grind — Answer Evaluation

The Mains evaluator accepts either a typed answer or a photograph of a handwritten response, alongside the question text and GS paper identifier. The system routes each submission through one of two evaluation paths:

Typed answers are evaluated using a paper-specific AI rubric — five distinct system instructions covering GS1 through GS4 and Essay, each encoding the unique evaluation criteria and examiner expectations for that paper. The output is normalised to a consistent schema: overall score, score rationale, strengths, weaknesses, keyword analysis (present, missing, and bonus keywords), structural feedback, examiner verdict, topper-level model answer, and priority improvement actions.

Handwritten answers are processed via Gemini Vision's multimodal capability. The model first faithfully transcribes the handwritten content, then evaluates the transcription using the same paper-specific rubric — replicating the actual UPSC Mains examination workflow. Illegible submissions are identified and surfaced with a clear user-facing message rather than producing a silent incorrect evaluation.

USAGE POLICY

Mains evaluations are rate-limited to **10 requests per user per 24-hour window** (enforced server-side by user identity, not IP), protecting AI infrastructure capacity while providing ample daily practice volume for serious aspirants. Administrative accounts operate under an elevated threshold.

Prelims Grind & Spaced Repetition

The Prelims PYQ bank is organised by subject — covering Polity, Economy, History, Modern History, Geography, Environment & Ecology, Art & Culture, Science & Technology, Social Issues, and International Relations under GS, plus Comprehension, Mathematics, and Reasoning under CSAT. Mains PYQs for GS1–GS4 are included for cross-stage topic synthesis. Each question entry includes the answer, a detailed explanation, topic classification, difficulty rating, and the year of appearance.

Attempt records are persisted server-side, enabling reliable cross-device accuracy analytics and profile-level performance statistics. Questions identified as difficult are added to a spaced repetition queue that schedules future review intervals using a simplified SM-2-style algorithm, promoting efficient long-term retention.

MCQ Test Series & Scoring Engine

The test series module delivers timed 100-question mock examinations with a question palette, live countdown, and review mode. On submission, all scoring is computed server-side at full floating-point precision using the exact UPSC marking scheme (2 marks correct, $-\frac{2}{3}$ marks wrong), ensuring that reported scores are trustworthy and free from the rounding drift common in client-side implementations. Derived performance metrics — accuracy, percentage, and performance band — are computed and persisted alongside the raw attempt data.

Each submission triggers an AI diagnostic report: a topic-wise breakdown is analysed to identify areas of underperformance, which are translated into a 7-day targeted study plan, a set of priority actions, and automatic population of the spaced repetition queue with the weakest topics. Past attempts may be re-analysed on demand, enabling aspirants to track whether targeted intervention produces measurable improvement.

Resource Library

The resource library provides curated, filterable access to NCERT books (with per-book completion tracking), reference books and topper notes organised by subject and module, and relevant YouTube class links. This reference layer complements the platform's active practice features and reduces the need to navigate external sources during a study session.

Authentication & Security

The platform supports email/password registration with industry-standard password hashing (bcrypt, cost factor 12) and Google OAuth via the Google Identity Services SDK. Both flows issue JWT-based sessions with configurable expiry. Visitor analytics use SHA-256 hashed IP addresses, ensuring that usage telemetry is privacy-safe and never exposes raw identifiers. Administrative access is gated by a role-based middleware chain separate from the standard user authentication flow.

Admin Panel — Behavioral Analytics

The administrative interface provides aggregate visitor analytics — daily request volumes, distinct visitor counts, and route-level breakdowns — drawn from hashed-IP log records. Administrators can manage the full user list and control platform-level feature flags, allowing features to be enabled or disabled dynamically without requiring a new deployment.

Beyond traffic reporting, the panel constructs full user-journey and session timelines using a 30-minute inactivity-gap heuristic to delineate discrete sessions, and tracks engagement across **9 named platform features**. Users are automatically segmented into retention cohorts — alongside feature discovery analysis and auto-generated narrative insights that translate raw event data into plain-language observations. Results are served through an in-memory cache with a 5-minute TTL to keep the dashboard responsive under repeated queries.

A companion **AdminStudyAnalytics** view aggregates subject-session analytics across the entire user base, surfacing subject-level study-intensity trends independent of any single aspirant's dashboard. The full behavioral analytics surface is gated behind combined `protect + adminOnly` middleware.

Multi-Provider AI Architecture

Designed with a multi-provider AI architecture that prioritises reliability and service continuity. The cascading provider chain (Gemini → Groq → OpenRouter with model rotation) keeps AI features operational even when individual providers experience outages or capacity constraints.

Server-Side Trusted Scoring

The MCQ scoring engine computes all results server-side at full arithmetic precision using the exact UPSC marking fraction. Client-submitted scores are never accepted, eliminating the score drift that is common in client-side implementations and ensuring that reported performance is fully trustworthy.

Persistent Student Memory

Background memory distillation continuously builds a curated model of each student's weak topics, study patterns, and exam goals from conversation history. This context is silently injected into future AI interactions, enabling genuinely personalised guidance without requiring the student to reintroduce themselves in each session.

Real-Time Cross-Device Sync

A WebSocket-based timer relay ensures that study time is accurately captured and reflected across all of a user's open tabs and devices simultaneously. The server is the authoritative source of truth, preventing both double-counting and data loss on connection changes.

Paper-Specific AI Evaluation Rubrics

Five distinct evaluation system instructions — one per UPSC GS paper and Essay — encode the unique examiner expectations and scoring criteria for each paper. GS4 emphasises ethical case-study analysis; GS1 emphasises historical and cultural depth; Essay emphasises thesis clarity and argumentative coherence. This level of specificity meaningfully exceeds generic AI evaluation tools.

Handwritten Answer Evaluation

Gemini Vision's multimodal capability enables aspirants to photograph handwritten practice answers for AI evaluation — a workflow that mirrors actual UPSC Mains examination conditions. The model transcribes and evaluates in a single pass, avoiding the latency and error propagation of separate OCR services.

Comparative Platform Analysis

Capability	UPSC Mentor	Typical Coaching App	Generic AI Chatbot
Multi-paper AI evaluation rubric	✓ 5 paper-specific	✗ Generic feedback	✗ No UPSC context
Server-side scoring (exact fraction)	✓ Full precision	Partial (rounded)	✗ Not applicable
Durable AI memory per student	✓ Background extraction	✗ Stateless	✗ Session-scoped
Cross-device real-time timer sync	✓ WebSocket relay	Polling-based	✗ Not applicable
AI-generated 7-day study plan	✓ Per test attempt	✗ Manual/generic	Ad hoc only
Handwritten answer evaluation	✓ Vision AI enabled	✗ Not available	Limited

The Aspirant's Journey — Supporting the Preparation Lifecycle

UPSC preparation typically spans 12–36 months and progresses through distinct phases. UPSC Mentor's feature architecture is deliberately aligned to each phase of this lifecycle, providing the right tools at the right moment rather than offering an undifferentiated toolkit.

1

Phase 1 — Planning & Baseline Setup

On registration, the aspirant configures their target exam date, daily study hour goal, and preparation year. The platform provisions all 48 syllabus modules and immediately renders the exam countdown, streak tracker, and a complete syllabus overview. The aspirant uses the Syllabus Tracker to survey the full GS Prelims and Mains structure — each module displays its official UPSC-sourced topic list — and begins marking areas of focus as study coverage commences.

2

Phase 2 — Daily Discipline & Execution

The Dashboard provides the operational command centre for daily preparation. The study timer resumes from the last server-persisted state and broadcasts live to all connected devices. The Today Planner structures daily goals; the Current Affairs Log captures the day's news. Study progress is charted in real time against the aspirant's target. Consecutive daily engagement increments the streak counter, gamifying consistency. The AI Mentor workspace is available throughout for concept clarification, current affairs interpretation, and strategy — informed by its accumulated memory of the student's preparation context.

3

Phase 3 — Active Testing & Diagnostic Intervention

As subject coverage deepens, the aspirant moves into active Prelims practice via the Prelims Grind module, working through subject-wise PYQs with detailed explanations. Difficult questions are marked for spaced repetition. The Topic-wise view cross-references Prelims and Mains PYQs, enabling comprehensive treatment of each topic across both examination stages. Timed mock tests with server-side scoring provide realistic assessment, followed by AI diagnostic reports that identify weak topics, prescribe a 7-day recovery plan, and populate the revision queue for targeted follow-up.

4

Phase 4 — Subjective Mastery & Iteration

With Prelims preparation consolidated, the Mains Grind becomes the primary tool. The aspirant selects a GS paper, submits a typed or handwritten answer, and receives an immediate structured evaluation: score, score rationale, keyword analysis (present, missing, and value-add terms), structural feedback across introduction, body, and conclusion, an examiner verdict, and a complete topper-level model answer demonstrating the standard to aspire to. Priority actions guide targeted improvement for the next attempt. The Answer Writing Tracker logs each submission for long-term progress review, and the AI Mentor supports this phase with nuanced contextual guidance — ethical case study framing for GS4, essay thesis development, outline review before drafting.

END-TO-END PREPARATION IN ONE PLATFORM

UPSC Mentor is architected so that an aspirant can progress through their entire preparation lifecycle — from day one through the eve of examination — within a single, unified environment. The shared data model means that weak topics surfaced in a mock test automatically appear in the revision queue, and the AI Mentor’s accumulated memory of those gaps informs the guidance it provides during Mains answer discussions. No context is lost between modules.

SECTION 07

Future Roadmap

UPSC Mentor's product roadmap is guided by the goal of deepening the platform's value across the full preparation lifecycle — from structured study planning through final interview readiness. The following initiatives represent the next phase of platform development, prioritised by user impact and strategic positioning.

Expanded AI-Driven Study Planning

Intelligent, adaptive study schedules generated from the student's syllabus coverage state, test performance history, and remaining time to examination — updated dynamically as the aspirant progresses.

Advanced Analytics Dashboard

A comprehensive performance analytics layer surfacing preparation velocity, subject-wise accuracy trends, answer writing improvement curves, and predictive readiness indicators across all platform features.

Personalised Revision Scheduling

Enhanced spaced repetition algorithms incorporating performance data from both MCQ practice and Mains evaluation to generate optimised, personalised revision calendars for each aspirant.

Mobile App Enhancements

Expanded native mobile capabilities including offline content access, push notifications for study reminders and streak alerts, and optimised mobile-first interfaces for on-the-go practice.

Community Learning Features

Peer discussion forums, answer-writing peer review, and collaborative study group tools — enabling aspirants to learn from each other's preparation strategies and build accountability networks.

Interview Preparation Support

AI-assisted mock interview simulations covering personality tests, structured DAF-based questioning, and current affairs discussion — extending the platform's utility through the final stage of the Civil Services selection process.

DEVELOPMENT PHILOSOPHY

Each roadmap initiative is designed to deepen the platform's core value proposition — reducing fragmentation, improving feedback quality, and enabling smarter, more personalised preparation — rather than expanding scope for its own sake. Features will be released iteratively with continuous validation against aspirant usage data and feedback.